Light

GCSE Computer Science – Computational Thinking (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is decomposition in computational thinking?

- ○ **A.** Breaking down a complex problem into smaller, more manageable parts
- ○ **B.** Making a program run faster
- ○ **C.** Removing unnecessary code from a program
- ○ **D.** Converting code from one programming language to another

Q2 (1 mark)

What is abstraction in computational thinking?

- ○ **A.** Adding more detail to make a solution complete
- ○ **B.** The process of removing unnecessary detail to focus on the important aspects of a problem
- ○ **C.** Creating abstract art using a computer
- ○ **D.** Writing code that nobody can understand

Q3 (1 mark)

What is algorithmic thinking?

- ○ **A.** Thinking about how to make computers work
- ○ **B.** Learning how to type quickly on a keyboard
- ○ **C.** Memorising complex mathematical formulas
- ○ **D.** Developing a step-by-step solution to solve a problem

Q4 (1 mark)

What is pattern recognition in computational thinking?

- ○ **A.** Finding similarities or common features in problems that can help create a general solution
- ○ **B.** Recognising shapes in images
- ○ **C.** Learning to recognise computer components
- ○ **D.** Identifying different programming languages

Q5 (1 mark)

In the context of computational thinking, what are inputs?

- ○ **A.** Data that is entered into a system for processing
- ○ **B.** The results produced by a program
- ○ **C.** The instructions that tell the computer what to do
- ○ **D.** The storage devices used by a computer

Q6 (1 mark)

Which of the following best describes a 'process' in a computer system?

- ○ **A.** Data waiting to be used
- ○ **B.** The final result displayed on screen
- ○ **C.** Actions performed on data to transform it from input to output
- ○ **D.** The keyboard used to type information

Q7 (1 mark)

What is the purpose of a structure diagram?

- ○ **A.** To show the physical layout of computer hardware
- ○ **B.** To design the user interface of a program
- ○ **C.** To draw flowcharts for programming

- ○ **D.** To display how a problem can be broken down into sub-problems in a hierarchical manner

Q8 (1 mark)

In a flowchart, what shape represents a decision?

- ○ **A.** Oval
- ○ **B.** Rectangle
- ○ **C.** Diamond
- ○ **D.** Parallelogram

Q9 (1 mark)

In a flowchart, what does a parallelogram symbol represent?

- ○ **A.** A process or calculation
- ○ **B.** The start or end of the program
- ○ **C.** A decision point
- ○ **D.** An input or output operation

Q10 (1 mark)

What is pseudocode?

- ○ **A.** A specific programming language used by professionals

- ○ **B.** A type of encryption used to protect programs
- ○ **C.** Code that has errors in it
- ○ **D.** A way of expressing an algorithm using structured natural language that resembles code but is not tied to any specific programming language

Q11 (1 mark)

What must every algorithm have?

- ○ **A.** Graphics and animations
- ○ **B.** A clear start point, a sequence of steps, and an end point
- ○ **C.** At least 100 lines of code
- ○ **D.** A graphical user interface

Q12 (1 mark)

A student needs to plan a program to manage a school library. Which computational thinking technique should they use first?

- ○ **A.** Decompose the problem into smaller sub-problems such as borrowing books, returning books, and searching for books
- ○ **B.** Write the code immediately
- ○ **C.** Design the graphics for the user interface
- ○ **D.** Choose which programming language to use

Q13 (2 marks)

A game developer is creating a driving simulation. They decide to focus on steering, acceleration, and braking, but ignore factors like individual engine components, the exact chemical composition of fuel, and air molecule movements.
Which computational thinking technique is being applied?

- ○ **A.** Pattern recognition, because they are looking for similar features
- ○ **B.** Decomposition, because they are breaking the problem into parts
- ○ **C.** Abstraction, because they are removing unnecessary detail to focus on the essential features of driving
- ○ **D.** Algorithmic thinking, because they are creating step-by-step instructions

Q14 (2 marks)

A programmer is asked to create a quiz application. They identify the following components: display question, get user answer, check answer, update score, display final result.
What have they produced through decomposition?

- ○ **A.** A complete working program
- ○ **B.** A debugging report
- ○ **C.** The final output of the program
- ○ **D.** A list of sub-problems that together form the whole solution

Q15 (2 marks)

A flowchart shows the following sequence: Start → Input num → Is num > 0? → (Yes) Output "Positive" → End. The (No) path leads to Output

"Not positive" → End.
What type of programming construct does the diamond shape represent?

- ○ **A.** Iteration, because it repeats the process
- ○ **B.** Selection, because the program chooses between two different paths based on a condition
- ○ **C.** Sequence, because the steps follow one after another
- ○ **D.** Assignment, because a value is stored in a variable

Q16 (2 marks)

Consider the following pseudocode:

```
total = 0
FOR count = 1 TO 5
    total = total + count
ENDFOR
OUTPUT total
```

What value will be output?

- ○ **A.** 10
- ○ **B.** 5
- ○ **C.** 15
- ○ **D.** 25

Q17 (2 marks)

A programmer notices that their code for calculating the area of a rectangle is very similar to their code for calculating the area of a floor in

a room, and also similar to their code for calculating the area of a computer screen.
What should the programmer do based on pattern recognition?

- ○ **A.** Create a single reusable function that calculates the area of any rectangle, and call it whenever needed
- ○ **B.** Delete all three pieces of code as they are duplicates
- ○ **C.** Keep all three separate pieces of code for clarity
- ○ **D.** Combine all the code into one very long program

Q18 (2 marks)

A program asks a user to enter their height in centimetres, converts it to metres by dividing by 100, then displays the result on screen.
Identify the input, process, and output.

- ○ **A.** Input: the display screen; Process: typing; Output: centimetres
- ○ **B.** Input: height in centimetres; Process: divide by 100; Output: height in metres displayed on screen
- ○ **C.** Input: metres; Process: multiply by 100; Output: centimetres
- ○ **D.** Input: the keyboard; Process: the screen; Output: the calculation

Q19 (2 marks)

In a flowchart, how is iteration (a loop) typically shown?

- ○ **A.** A single arrow pointing downward to the next step
- ○ **B.** A dotted line between two separate flowcharts

- ○ **C.** A double-lined rectangle around multiple steps
- ○ **D.** An arrow that leads back to an earlier point in the flowchart, creating a cycle

Q20 (2 marks)

Consider the following algorithm:

```
INPUT number
IF number MOD 2 == 0 THEN
    OUTPUT "Even"
ELSE
    OUTPUT "Odd"
ENDIF
```

What is the purpose of this algorithm?

- ○ **A.** To double a number
- ○ **B.** To determine whether a number is even or odd
- ○ **C.** To divide a number by 2
- ○ **D.** To find the remainder when dividing by 2

Q21 (2 marks)

In a structure diagram for an online shopping system, 'Checkout' is broken down into 'Enter delivery address', 'Choose payment method', and 'Confirm order'.
What is the relationship between 'Checkout' and these three components?

- ○ **A.** They are all at the same level and have no relationship

○ **B.** 'Checkout' is the parent and the three components are its children, showing hierarchical decomposition

○ **C.** The three components are parents of 'Checkout'

○ **D.** 'Checkout' and the components are alternatives that users choose between

Q22 (2 marks)

Why is computational thinking important when developing software?

○ **A.** It makes programs run faster on any computer

○ **B.** It eliminates the need for testing

○ **C.** It automatically generates code without human input

○ **D.** It helps programmers understand, analyse, and solve complex problems systematically before writing code

Q23 (2 marks)

Consider the following pseudocode:

```
x = 10
y = 3
z = x DIV y
r = x MOD y
OUTPUT z
OUTPUT r
```

What values will be output?

○ **A.** z = 3.33, r = 0

- ○ **B.** z = 3, r = 1
- ○ **C.** z = 30, r = 10
- ○ **D.** z = 1, r = 3

Q24 (3 marks)

A software company is developing a weather forecasting application. The project manager asks the team to use both decomposition and abstraction during the planning phase.
Explain how both techniques would be applied and why each is important.

- ○ **A.** Decomposition and abstraction are the same thing — both involve breaking the problem into smaller parts
- ○ **B.** Decomposition breaks the weather app into components (data collection, processing, display, alerts); abstraction removes irrelevant atmospheric details to focus on key factors like temperature, pressure, and humidity. Decomposition makes the project manageable; abstraction makes the model practical to implement.
- ○ **C.** Abstraction breaks the problem into parts while decomposition removes unnecessary details
- ○ **D.** Neither technique is relevant to weather application development

Q25 (3 marks)

A flowchart represents the following algorithm:

```
Start → INPUT password → Is password == "secret123"? → (Yes) OUTP
                                                      → (No) OUTPUT
```

Analyse this algorithm and identify its weaknesses.

- ○ **A.** The algorithm is perfect and has no weaknesses
- ○ **B.** The only weakness is that it uses a flowchart instead of pseudocode
- ○ **C.** Weaknesses include: the password is hardcoded making it impossible to change without modifying code; there's only one attempt allowed with no retry option; the password is stored as plain text which is insecure; there's no account lockout after failed attempts; the simple comparison could be vulnerable to timing attacks
- ○ **D.** The weakness is that it outputs text messages instead of numbers

Q26 (3 marks)

Consider the following pseudocode:

```
array numbers = [4, 2, 7, 1, 9]
max = numbers[0]
FOR i = 1 TO 4
    IF numbers[i] > max THEN
        max = numbers[i]
    ENDIF
ENDFOR
OUTPUT max
```

Analyse this algorithm: what is its purpose and what will be output?

- ○ **A.** Purpose: sort the array; Output: 1
- ○ **B.** Purpose: find the minimum value in the array; Output: 1
- ○ **C.** Purpose: find the maximum value in the array; Output: 9
- ○ **D.** Purpose: calculate the sum of the array; Output: 23

Q27 (3 marks)

A student is designing a structure diagram for a mobile banking application. They have identified the main system and need to show the first level of decomposition.
Which breakdown would be most appropriate for the first level?

- ○ **A.** Login, View Balance, Transfer Money, Pay Bills, View Statements, Settings
- ○ **B.** Enter Username, Enter Password, Click Button, Wait for Response
- ○ **C.** Start App, Use App, Close App
- ○ **D.** Screen 1, Screen 2, Screen 3, Screen 4, Screen 5

Q28 (3 marks)

A programmer has written separate algorithms to calculate:

- The average of student test scores
- The average temperature over a week
- The average speed during a journey

Using pattern recognition, how should the programmer improve their approach?

- ○ **A.** Keep all three separate algorithms as they deal with different data
- ○ **B.** Delete two of the algorithms and only use one scenario
- ○ **C.** Recognise the common pattern (sum all values, divide by count) and create a single generalised 'calculate average' function that accepts any list of numbers as a parameter, then call this function for each scenario
- ○ **D.** Combine all three into one algorithm that always calculates all three averages together

Q29 (3 marks)

A teacher asks students to represent an algorithm that validates a password (must be at least 8 characters and contain at least one number).
Compare the suitability of flowcharts versus pseudocode for this task.

- ○ **A.** Flowcharts are always better because they are visual
- ○ **B.** Pseudocode is always better because it looks like code
- ○ **C.** Flowcharts provide a clear visual representation of the decision logic and flow, making them ideal for planning and communication; pseudocode is closer to actual code syntax, making it easier to translate into a programming language. Both are valid representations with different strengths depending on the purpose.
- ○ **D.** Neither flowcharts nor pseudocode can represent password validation

Q30 (3 marks)

A sat-nav system needs to calculate the fastest route between two locations. Different levels of abstraction could be applied to this problem.
Analyse how different levels of abstraction would affect the system.

- ○ **A.** Abstraction is not relevant to route-finding problems
- ○ **B.** Only one level of abstraction should ever be used
- ○ **C.** High abstraction (simple distance calculation): faster processing but ignores traffic, road types, and speed limits, giving inaccurate times. Medium abstraction (road network with speed limits): balances accuracy and performance by modelling key factors. Low abstraction (every traffic light, pedestrian crossing, road surface): extremely accurate but too slow for real-time use and requires excessive data. The appropriate level depends on the required balance between accuracy and performance.

○ **D.** Lower abstraction always gives better results

Total: **55** marks • Time: **30 min**

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